AMD DEVELOPER HACKATHON · ACT III · INTELLIGENT INDUSTRY

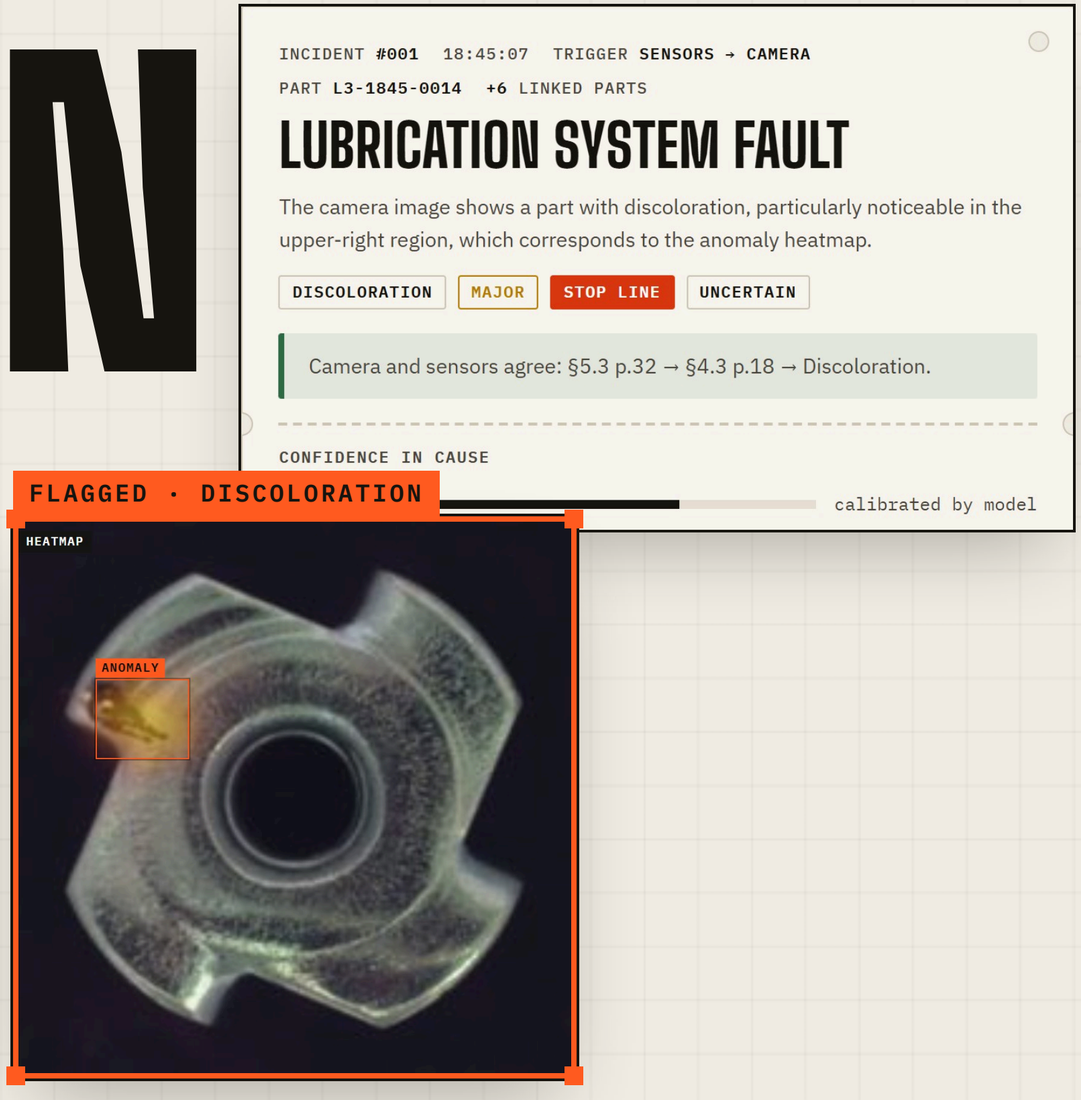
FOREMAN

FIND IT. EXPLAIN IT. FIX IT.

A shop-floor inspection agent. PatchCore finds the defect, Gemma 4 explains why, the machine manual says how to fix it, and a supervisor signs off.

ONE AMD RYZEN CPU | **NO GPU · NO CLOUD**

github.com/szae444/foreman-amd-hackathon



A WORN DIE COSTS YOU TWICE.

01

SCRAP

Bad parts keep coming off the line until someone notices.

02

DOWNTIME

A technician still has to work out **why**. The answer is in a thick manual and in senior technicians' heads.

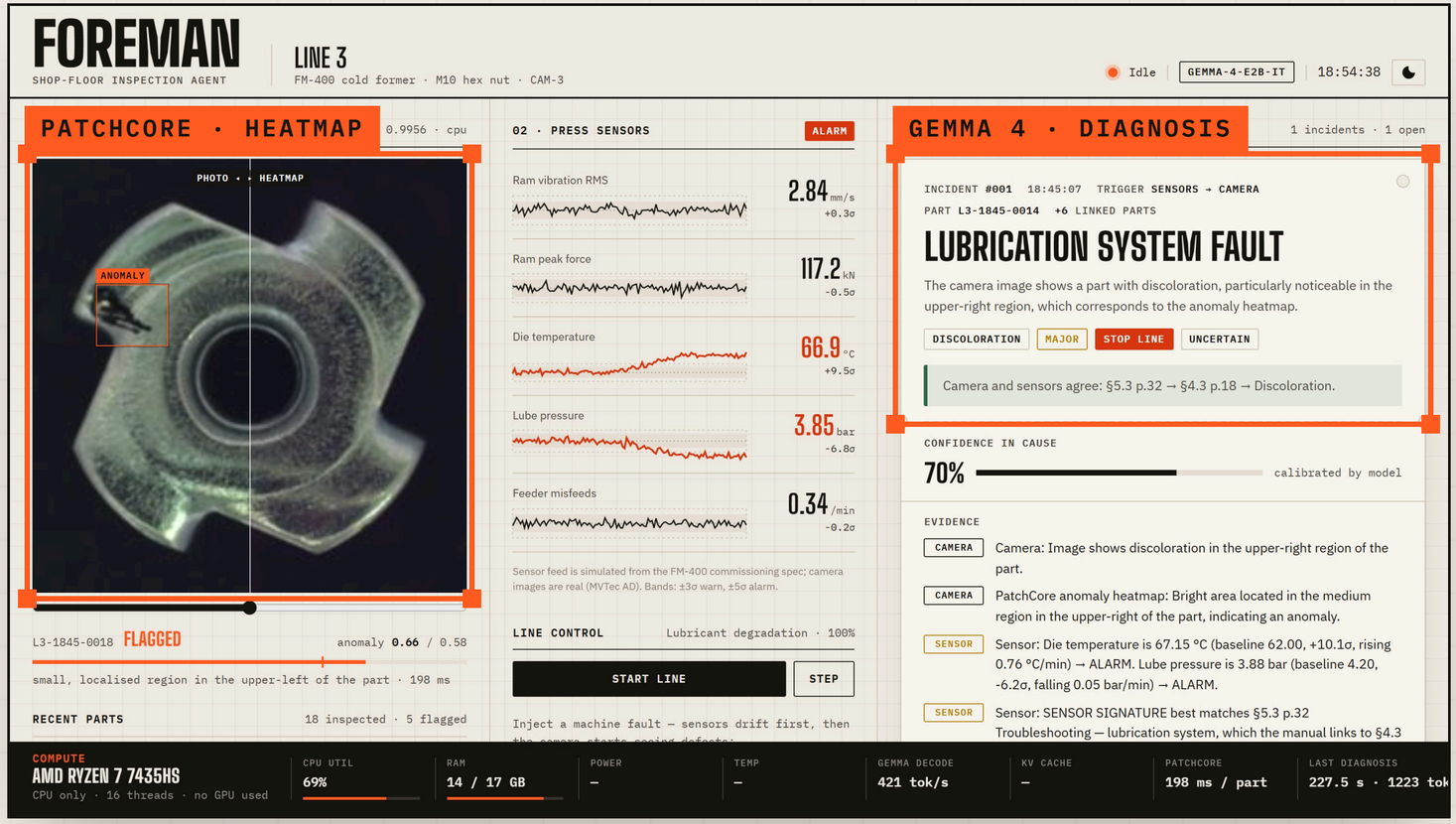
03

BLIND SPOTS

Vision systems say "bad part", not why. And cloud AI can't see a camera feed that has to stay in the plant.

Symptom → cause → fix lives in a manual and in experienced people

FROM CAMERA IMAGE TO APPROVED WORK ORDER.



Real output

This is the live app during the lubricant fault, with Gemma 4 E2B running CPU-only on the AMD Ryzen.

The camera flags a discoloured nut, the heatmap boxes the stain, and Gemma names a **lubrication system fault**.

Foreman's own check confirms camera and sensors agree: manual §5.3 → §4.3.

Camera images: real (MVTec AD) · press sensors: simulated

SIX STEPS. TWO ARE NEURAL.

GEMMA 4 E2B · LLAMA.CPP

01
DETECT
PatchCore, trained on good parts only, scores each part and draws the heatmap.

02
FUSE
Press sensors against the machine's baseline. Statistics, not an LLM.

03
RETRIEVE
Manual sections that match the signature, plus the repairs they point to.

04
REASON
Gemma 4 vision reads photo, heatmap, sensors and manual. Structured diagnosis.

05
VERIFY
Every manual citation is checked against what the model was shown.

06
APPROVE
A supervisor approves or rejects. The repair runs, the sensors recover.

■ Dark steps run on the AMD Ryzen CPU: PatchCore on PyTorch, Gemma 4 on llama.cpp.

SENSORS WARN FIRST. THE CAMERA CONFIRMS.

1 · Fault starts

Lubricant degrades. The parts still look fine.

2 · Early warning

Die temperature up, lube pressure down. Foreman opens an incident from the sensors alone.

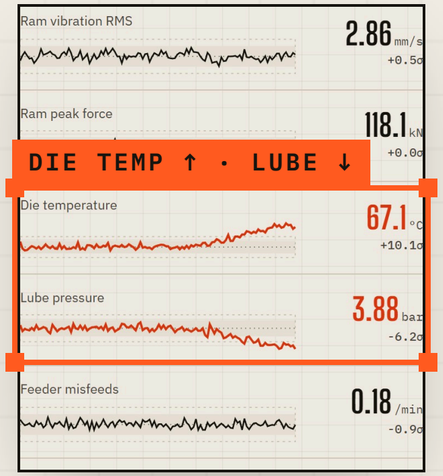
3 · Escalated

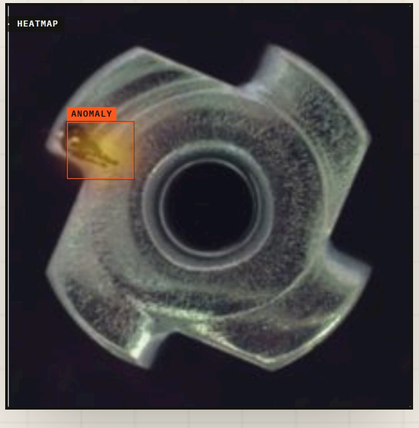
The camera sees discoloration. The **same** incident is re-diagnosed with the photo.

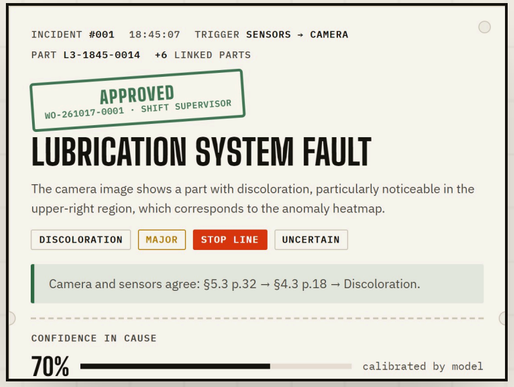
4 · Approved

Filter change per the manual. The sensor traces return to baseline.

ONE INCIDENT. ONE WORK ORDER. NO ALERT STORM.







ONE RYZEN CPU. TWO MODELS.
NO GPU. NO CLOUD.

THE MACHINE · LIVE IN THE UI

COMPUTE
AMD RYZEN 7 7435HS
CPU only · 16 threads · no GPU used

CPU UTIL	RAM	POWER	TEMP
22%	15 / 17 GB	—	—

0
GPUs used. The laptop's NVIDIA card is switched off for the run.

184 ms
PatchCore per part on the Ryzen 7 7435HS

12 tok/s
Gemma 4 E2B decode on the same 8 cores

12/12
live eval: right defect, agreement and § cited

MEASURED, NOT CLAIMED.

DETECTOR

0.996

image AUROC · MVTec AD metal_nut
held-out test split

MEASURE	AMD RYZEN (CPU)	RTX 4050 (DEV)
False alarms on good parts	0 %	same model
Recall: bent / color / flip / scratch	96/96/100/78 %	same model
PatchCore latency per part	184 ms	—
Gemma 4 E2B full diagnosis (median)	2.8 min	≈ 18 s
Live eval: right defect + § cited (4 faults)	12 / 12	12 / 12
Repair matches the manual	9 / 12	—
Automated tests	34	—

The 3 repairs that missed were flagged by Foreman's own checks. Small sample: 3 images per fault.

FROM ONE DEMO LINE TO A REAL PLANT FLOOR.

Your manuals

Any manual in the section format. Cross-references are followed, citations stay checked.

Your sensors

Swap the simulator for the plant's historian or an OPC UA feed.

Your CMMS

Approved work orders go straight into the maintenance system, with the evidence attached.

FIND IT. EXPLAIN IT. FIX IT.